

LBS-IntentBench: A Real-World Benchmark for Implicit Intent Inference and Spatio-temporal Reasoning

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Abstract

Large Language Models (LLMs) have advanced recommendation systems, yet their application in Location-Based Services (LBS) remains challenging because user intent is tightly coupled with spatio-temporal context and physical constraints. Existing benchmarks mainly rely on explicit instructions or static text matching, and therefore fail to evaluate implicit intent understanding from real-world mobility behaviors. To address this gap, we propose LBS-IntentBench, a benchmark built on large-scale anonymized real-world user trajectories. It evaluates LLMs through three complementary modules: mobility intent inference, which ranks latent user needs from historical behaviors and contextual signals; contextual constraint inference, which identifies candidate intents that jointly satisfy temporal, spatial and causal constraints; and general mobility tasks, which assess capabilities such as POI understanding, next-step prediction, and counterfactual analysis. We benchmark a range of state-of-the-art LLMs on LBS-IntentBench, providing a standardized diagnostic framework for assessing model capabilities in real-world mobility scenarios. Our benchmark fills an important gap in evaluating recommendation systems beyond passive ranking and offers a realistic testbed for future research on context-aware and physically grounded LLM-based recommenders. The benchmark is provided in <https://github.com/lbs-researcher/LBS-IntentBench>.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Intent Inference, Spatio-temporal Reasoning, Benchmark, Large Language Models, Location-Based Services

1 Introduction

Large language models (LLMs) are reshaping recommender systems from passive feature-matching ranking [13, 15] into generative decision-making grounded in semantic understanding and multi-hop reasoning [14, 17, 28, 33]. Nowhere is this shift more consequential than in location-based services (LBS). Unlike e-commerce or content recommendation, which are dominated by long-horizon preference modelling [25, 35], LBS decisions are inseparable from spatio-temporal context, physical feasibility and real-time environment [16, 18, 30]. This dependency exposes three capabilities that sequential and collaborative-filtering paradigms [13, 15] cannot natively support: (i) implicit-intent reasoning over fragmented navigation, click and search signals; (ii) multi-constraint plausibility reasoning that jointly arbitrates temporal, spatial, meteorological, accessibility and personal constraints; and (iii) structured cognition across long-horizon trajectories, spanning fact extraction, pattern induction and counterfactual verification. These demands mirror precisely the comparative advantages of LLMs—and the sharpest blind spots of incumbent LBS paradigms.

Consider a canonical case. At 18:40 on a rainy Friday in Shanghai, a user steps out of the office without issuing a query. Their recent signals reveal a 19:00 dinner reservation in Xujiahui, a 20:00 cinema booking nearby, and a six-week pattern of Friday grocery stops at a fixed supermarket on the commute home. Conventional LBS systems, deprived of an explicit intent, fall back to the home route or nearby popular POIs—neither aligned with the user’s true state. An LLM-driven system should instead infer a stack of concurrent intents: the primary intent is to depart for Xujiahui now, with rain-induced travel latency tightening the 19:00 deadline; the secondary grocery routine must be actively suppressed, as it no longer fits the window. Such decisions hinge on implicit-intent discovery, causal multi-constraint ranking and time-window feasibility reasoning. They recur at massive daily scale on real LBS platforms, yet no existing benchmark characterizes them systematically.

Prior work has begun to probe this space. Benchmarks such as TravelPlanner [29], TravelBench [6] and MobilityBench [24] report encouraging progress on explicit tasks—itinerary planning, route generation—under multi-constraint settings. However, two assumptions limit their external validity: tasks are predicated on an explicit natural-language instruction, and are often synthesized from static text decoupled from real mobility trajectories. In production LBS, by contrast, user needs are fragmented, implicit and temporally evolving; the system must actively infer latent intents from raw behavioural trajectories and commit to decisions that simultaneously satisfy spatio-temporal and physical constraints. Current evaluation paradigms bypass this perception-to-decision chain, producing a pronounced offline-deployment effectiveness gap.

To close this gap, we present LBS-IntentBench, a benchmark dedicated to implicit mobility intent in LBS recommendation, constructed from large-scale, de-identified real-user trajectories. The benchmark is organized around an Intent–Decision–Fact hierarchy that decomposes LBS decision-making into three mutually reinforcing layers. Mobility Intent Inference (MII) probes the discovery of user intent from implicit behavioural signals in the absence of any explicit query. Contextual Constraint Inference (CCI) stresses the identification of candidate intents that jointly satisfy temporal, spatial and causal constraints. General Mobility Tasks (GMT) evaluates spatio-temporal factual cognition—from fine-grained entity recognition to macroscopic behavioural induction—and provides the diagnostic substrate on which intent and constraint reasoning rest. Together, the three tasks enable both capability-level ablation and end-to-end diagnosis of the perception-to-decision chain.

Our contributions are summarized as follows:

- We propose *LBS-IntentBench*, a benchmark built on large-scale real-world mobility trajectories specifically designed for implicit intent understanding and spatio-temporal reasoning in LBS recommendation.

- We design a multi-level evaluation framework providing a standardized tool for diagnosing model capabilities in real-world decision-making settings.
- We benchmark a diverse set of LLMs, and show that current models remain far from reliable in implicit intent ranking and multi-constraint mobility reasoning despite strong performance on local factual tasks.

2 Related Work

2.1 LLMs for Recommendation Systems

Large language models are reshaping the modeling paradigm of recommender systems [17, 28]. Early work such as P5 [9] reformulated recommendation as a text-to-text generation problem, while TALLRec [4] further improved the adaptation of LLMs to recommendation tasks through recommendation-data fine-tuning, showing that LLMs can be efficiently aligned to recommendation objectives even with limited domain-specific data. More recently, OpenOneRec [34] proposed a unified training framework for recommender systems and introduced RecIF-Bench to systematically evaluate model capabilities in basic prediction, instruction following, and complex reasoning. In LBS scenarios, CoAST [32] incorporates world knowledge and spatio-temporal trajectories through continued pre-training and cognitive alignment for next POI prediction, while ROS [19] introduces hierarchical geographic semantic representations and spatial chain-of-thought reasoning to enable geography-aware LLM inference; however, these works mainly focus on explicit tasks such as next POI prediction, and they do not systematically evaluate whether models can identify and infer implicit intent from real mobility trajectories.

2.2 Spatio-temporal Planning and Agent Benchmarks

With the growing capability of LLMs, researchers have proposed a variety of benchmarks for complex tasks and spatio-temporal planning. In travel and spatial reasoning, TravelPlanner [29], TravelBench [6], TripCraft [5], and MobilityBench [24] evaluate model execution under real-world constraints through itinerary planning, multi-turn interaction, or route generation tasks. These benchmarks have significantly improved planning performance under complex conditions, but they mainly test the understanding and execution of explicit natural-language instructions, such as when a user directly specifies a destination, travel goal, or constraint. In contrast, real-world mobility behavior is often fragmented, implicit, and context-dependent; therefore, instruction-driven evaluation alone cannot cover the key capability of implicit intent understanding. Meanwhile, some work in the general domain has begun to study complex or implicit intent understanding [27], but such benchmarks typically lack real spatio-temporal context and physical constraints, making them difficult to transfer directly to LBS scenarios.

2.3 Intent and Semantic Understanding from Mobility Trajectories

Real-world mobility trajectories not only record users’ spatial movement, but also implicitly encode visiting intentions and travel preferences. Mobility-LLM [10] learns user intent and travel preferences directly from check-in sequences, indicating that trajectory semantics can be explicitly modeled. MobQA [3] further organizes mobility-data understanding into factual retrieval, multiple-choice reasoning, and free-form explanation tasks, showing that the spatio-temporal and semantic information in trajectories can support question-answering-style evaluation. However, these works either focus on method development or on a single type of semantic understanding task; they have not yet formed a unified benchmark that systematically evaluates whether models can infer implicit intent from real trajectories and perform consistent reasoning under physical constraints.

3 LBS-IntentBench Methodology

This section will detail the construction methodology of LBS-IntentBench. To comprehensively and systematically evaluate the capabilities of large language models in real-world mobility recommendation scenarios, we construct evaluation modules from three levels: "intent-decision-fact".

Task 1 (Mobility Intent Inference, MII) is used to evaluate whether the model can identify and rank users’ latent mobility needs from historical behaviors and spatio-temporal context; **Task 2 (Contextual Constraint Inference, CCI)** is used to evaluate whether the model can identify the current intent and decision logic based on contextual information; **Task 3 (General Mobility Tasks, GMT)** contains 7 sub-tasks across three levels: basic factual understanding (POI semantic understanding, trajectory fact retrieval), behavior sequence modeling (next-step mobility prediction, location-time reasoning), and mobility induction and reasoning (mobility preference summarization, mobility reason inference, counterfactual anomaly detection), progressively testing the model’s understanding and reasoning capabilities of the real world in mobility scenarios.

3.1 Basic Data

The data for this benchmark is strictly driven by desensitized real user behavior, and the following three types of datasets are directionally provided for different tasks:

User Intent Basic Data (for Task 1 and 2): The MII and CCI tasks are evaluated based on this basic data. Each sample includes:

- **User Profile (Static Profile):** Contains long-term stable attributes such as age group, gender, city of residence, family structure, consumption level, and whether they own a private car. This is used to depict the user’s relatively unchanging mobility background and capability boundaries.
- **Behavior Sequence:** Organized in reverse chronological order, retaining 20–30 raw behavior records prior to the trigger time. Each record contains: timestamp, POI name or category, industry category, behavior type (navigation/click/search, etc.), whether arrived, local/non-local

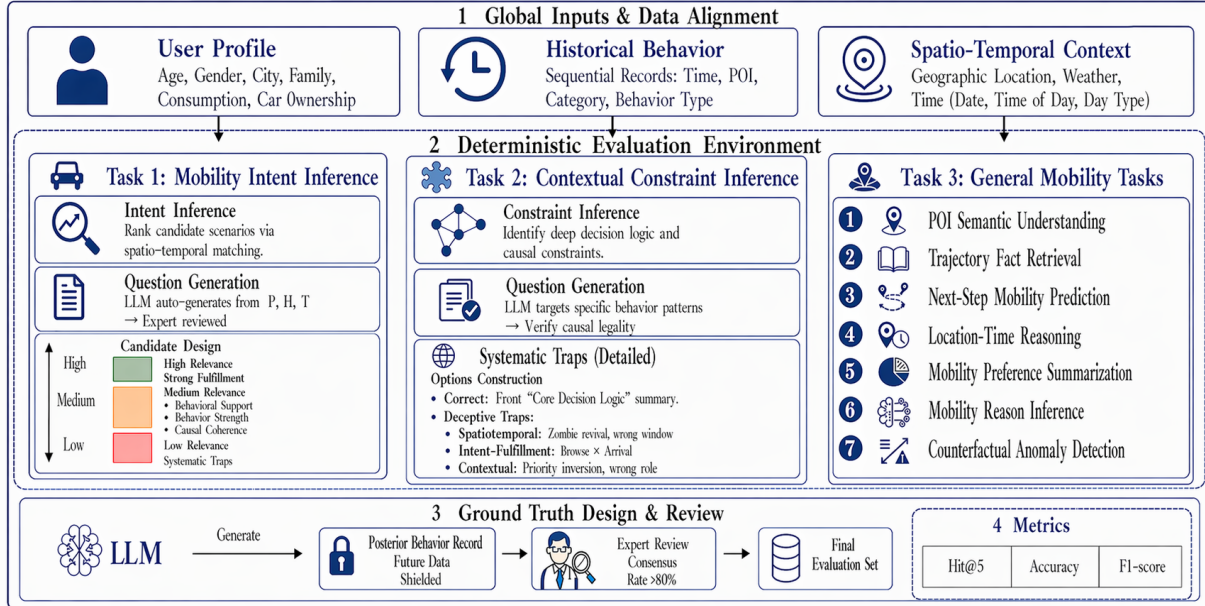


Figure 1: Overview of LBS-IntentBench. The benchmark evaluates LLMs from three complementary perspectives: mobility intent inference, contextual constraint inference, and general mobility tasks.

attribute, price segment, as well as holiday and meteorological information. To prevent shortcut learning, any intent intensity scores are strictly hidden in the samples, forcing the model to rely solely on observable behavior trajectories for inference. Among them, records with confirmed arrivals accompanied by navigation behaviors constitute strong intent signals, while records with only clicks or browsing serve as weak intent signals. The distinction between the two is the core basis for subsequent intent scenario attribution.

- **Spatio-temporal Trigger:** Includes precise trigger time (accurate to the minute), district-level geographic location, real-time weather, temperature, wind power, and other immediate environmental variables.

Mobility Trajectory Data (for Task 3): We align, deduplicate, and merge user behavior logs and geographic location stay records to form a unified chronological event sequence; each user trajectory retains a maximum of 150 chronologically ordered events. Events are divided into two categories: stay events and action events. Stay events record the user’s stay information at a certain POI, including the start and end time of the stay, POI name, and category tags; action events record the user’s transfer process, including behavior type, origin and destination, travel mode, trip distance, time consumed, whether arrived, local/non-local attribute, and weather information. For tasks involving future prediction, we set a truncation point in the latter half of the trajectory and remove events identical to the ground-truth POI within 15 minutes before the truncation point. Meanwhile, distractors are extracted from the filtered history to prevent information leakage.

Real-World POI Knowledge Base (for Task 3): To support POI semantic understanding tasks, we further constructed a high-quality POI knowledge base. This knowledge base filters records from internal POI data that simultaneously possess complete province, city, and district information, category tags, tag weights, and text introductions, and filters them based on the length of the introduction. To ensure semantic quality, we use an independent model (DeepSeek V3.2) to perform consistency judgments on each (POI, introduction) data pair, requiring that the introduction can correspond to the POI’s geographic location, surrounding landmarks, business attributes, or functional descriptions, and that there are no obvious factual contradictions or over-generalization issues. Ultimately, we retained 1,000 high-quality POI records as the evaluation set to support entity cognition and semantic understanding tasks.

3.2 Evaluation Protocol

To cover evaluation needs from basic perception to complex decision-making, LBS-IntentBench adopts three standardized paradigms: ranking, multiple-choice, and open question answering. These paradigms correspond to different interaction settings and are used to evaluate different model capabilities. The templates for all these tasks are presented in Table 1.

- **Ranking Paradigm:** Intents in real scenarios often possess polysemy and ambiguity. This question type requires the model to prioritize a set of potential intent scenarios.
- **Multiple-Choice Paradigm:** By systematically injecting deceptive distractor options (such as spatio-temporal violations, causal inversions), this standardizes the evaluation of the model’s discrimination and logical error-correction capabilities.

Table 1: Overview of Benchmark Tasks

Task	Format	Description
Mobility Intent Inference	Ranking	Rank candidate intent scenarios by likelihood given user profile, behavior history, and spatio-temporal context.
Contextual Constraint Inference	Multiple-choice	Identify behavioral constraints and decision logic from candidates given profile, history, and spatio-temporal context.
POI Semantic Understanding	Q&A	Given a POI name, predict its location (province/city/district) and category tags. Given a textual description, predict a ranked list of matching POI names.
Trajectory Fact Retrieval	Q&A	Identify the POI where the user is located at a specific timestamp from the trajectory. Retrieve all stay intervals at a specified POI from the trajectory. Identify the most interested POI and its frequency within a given time window.
Next-Step Mobility Prediction	Q&A	Predict the next destination POI name given a historical trajectory and the current spatio-temporal context. Predict the next destination POI tags given a historical trajectory and the current spatio-temporal context.
Location-Time Reasoning	Multiple-choice	Predict the most likely POI the user will visit at a given future timestamp. Predict the most likely start-end interval of the user's first stay at a given POI.
Mobility Preference Summarization	Q&A	Identify the user's dominant travel mode, given the trajectory. Identify the user's preferred travel mode when visiting a given destination category, given the trajectory.
Mobility Reason Inference	Multiple-choice	Infer underlying motivations for a travel event (time, destination, and travel mode) from a set of candidate reasons.
Counterfactual Anomaly Detection	Q&A	Given two shuffled trajectory snippets (one original, one with a subtly tampered event), judge which one is more reasonable. Locate the exact index of the anomalous event within the tampered snippet.

- **Open Question Answering:** Requires the model to directly extract facts from trajectories (such as time points, stay durations) and predict the next step. Because no option hints are provided, it can most authentically reflect the model's raw perception accuracy of spatio-temporal data and whether hallucinations exist.

Beyond the question format, LBS-IntentBench adopts task-specific evaluation metrics to capture different levels of difficulty and different aspects of model behavior. For ranking-based MII, we report both **Exact Match**, which requires the full candidate ranking to be correct, and **Top-1 Accuracy**, which only checks whether the highest-priority intent is correctly identified. We further report fine-grained accuracies by distractor type (e.g., temporal misalignment, profile constraint violation, and deep conflict) to diagnose which kinds of misleading signals are most challenging for models.

For multiple-choice CCI, we distinguish between **Single** and **Multi** settings, and report **Overall** accuracy as the sample-count-weighted average across the two. In addition, we report fine-grained exact-match accuracies by trap type, including spatio-temporal validity, intent-fulfillment confusion, and contextual attribution, so as to measure models' robustness under different kinds of constraint violations.

For GMT, we use metrics that match the output structure of each subtask: **Accuracy** for exact categorical or option prediction, **F1** for multi-label tag matching, **Hit@5** for POI candidate recall. This multi-metric design allows us to evaluate not only whether a model produces a plausible answer, but also whether it is fully correct under the compositional and structured requirements of real-world mobility reasoning.

3.3 Task 1: Mobility Intent Inference

3.3.1 Task Definition. User mobility intents in real LBS scenarios have inherent polysemy and probability: under the same spatio-temporal background, a user may simultaneously hold multiple reasonable potential needs, and their priorities are jointly modulated by historical behavior patterns, personal profile features, and

current spatio-temporal context. Existing evaluation systems generally adopt a "single correct answer" discriminative paradigm, which cannot objectively measure the model's global ranking ability for intent priorities.

Therefore, we define Task 1 as an **intent ranking task**: given a static user profile $\mathcal{P}$, historical behavior sequence $\mathcal{H}$, and current spatio-temporal trigger context $\mathcal{T}$, the model is required to output a priority ranking σ for a candidate scenario set $\mathcal{O} = \{o_1, o_2, \dots, o_n\}$ based on "the intent most likely to occur for the user at the current moment," which is then compared with expert-annotated ranking results.

3.3.2 Task Construction. This section introduces the overall construction process of the evaluation data. It includes three stages:

Step 1: Preliminary Data Preparation Transforming raw behavior sequences into evaluable structured scenario data. Table 2 is an example of this process.

- **Intent Attribution:** Using an "environment-evidence-motivation" triplet to conduct structured semantic annotation on users' raw behaviors, extracting spatio-temporal attributes, arrival/intent signals, and latent decision preferences.
- **Intent Aggregation:** Merging behaviors that have spatio-temporal continuity and serve the same goal into the same intent, and attaching a time status (execution phase/decision phase/completion phase) and interest heat annotation to each intent.

Step 2: Question Design (Candidate and Distractor Construction) The question includes: static user profile, historical behavior sequence, current spatio-temporal snapshot, and 3–4 candidate intent scenarios.

- **Candidate Design (Gradient Design):** All candidate scenarios use standardized semantic descriptions that strictly desensitize specific POIs. Based on spatio-temporal validity, behavior evidence strength, and causal coherence, candidates are divided into three relevance tiers: "High, Medium, and Low" (Relevance Tiers). This gradient design upgrades the evaluation from a simple binary discrimination to an

Table 2: Example of Intent Attribution and Scenario Aggregation

Component	Details
User Profile	Beijing user (Female, 30–34 years old, no car).
Spatio-temporal Trigger	Sunday 17:03, Location: C Street.
Relevant Records	SPA category (Today, Navigation, Arrived=1); Shopping·Stationery (Today, Route planning, Arrived=0); Bookstore/Art Supplies/Art District (3 dates in past 5 weeks, Click/Planning, Arrived=0).
Intent Attribution	<i>Environment:</i> Sunday afternoon, user completed SPA. <i>Behavior:</i> Initiated route planning for nearby stationery; historical cross-week occurrences ≥ 3 times. <i>Inference:</i> Creative arts priority; aesthetic driven > mass entertainment; purposeful mobility > wandering.
Scenario Aggregation	<i>Event:</i> "Local-Sunday SPA experience" (Status: Execution phase; Cycle: One-time). <i>Preference:</i> "Local-Weekend arts and cultural consumption" (Status: Execution phase; Cycle: Periodic; Trend: Stable).

ordinal Ranking Task, focusing on testing the model’s fine-grained discrimination and global preference ranking capabilities among multiple Competing Explanations.

- **Distractor Design (Systematic Traps):** Targeting common error paths of large models, three types of low-compliance distractors are systematically injected:
 - (1) **Temporal Misalignment Trap:** Dormant intents or time-misaligned intents, testing the ability to model temporal decay. (e.g., treating a dormant intent as a recent preference, or describing a weekend behavior as a workday one.)
 - (2) **Profile Constraint Violation Trap:** Generating intents that conflict with the user’s core attributes, testing the model’s awareness of profile-level constraints. (e.g., recommending parent-child entertainment to a single user.)
 - (3) **Causal Inversion Trap:** Treating already-completed intents as still upcoming, or fabricating intents that violate spatio-temporal common sense, testing the model’s deep causal reasoning ability. (e.g., recommending commuting to a user who has already gotten off work and is at home.)

Step 3: Reference Label Construction and Review Final reference labels are obtained through a two-stage adjudication pipeline. Each instance is first independently assessed by six LLM judges under a shared annotation protocol defining evidence interpretation rules and the distractor taxonomy. Their outputs are then aggregated in a second-round adjudication step for consistency checking

and label consolidation. Instances with unanimous agreement (6/6) are directly retained.

For high-consensus but non-unanimous cases (5/6 agreement), we invoke additional review by five domain experts. An instance is retained only when at least four of the five experts agree on the final annotation ($\geq 80\%$ agreement); otherwise, it is discarded or re-annotated. This design mitigates single-judge bias and yields more reliable benchmark labels.

3.4 Task 2: Contextual Constraint Inference

3.4.1 Task Definition. Users’ real decision-making behaviors are not simple mappings of isolated preferences, but a causal system jointly driven by spatio-temporal constraints, decision styles, and psychological trade-offs. Existing recommendation evaluation benchmarks generally simplify user understanding to behavior pattern matching, which is neither able to verify the attribution accuracy of "motivations behind behaviors," nor able to identify deep reasoning errors such as time window mismatches, intent-fulfillment confusion, and cross-role logical contamination. Therefore, we define Task 2 as a **Contextual Constraint Inference (CCI)** task: given a static user profile $\mathcal{P}$, historical behavior sequence $\mathcal{H}$, and spatio-temporal trigger information $\mathcal{T}$, the model is required to accurately identify the deep decision logic driving the user’s behavior from candidate options, covering three levels: the correctness of behavior motivation attribution, the physical validity of the spatio-temporal window, and the rationality of decision priority ranking.

3.4.2 Data Construction. The construction of Task 2 is divided into three steps.

Step 1: Preliminary Data Preparation. This step shares the same behavior evidence triplets and mobility intents as Task 1, but shifts the objective from intent prediction to behavior attribution and contextual constraint inference.

Step 2: Question Design. To comprehensively evaluate the reasoning robustness of the model, the evaluation tasks uniformly adopt a multiple-choice format containing 4–5 candidates, mixing single-choice and multiple-choice questions. The question stem provides the user’s multi-dimensional context (profile, historical sequence, spatio-temporal trigger signal), and the candidates are descriptions of the user’s potential decision logic.

Distractor design is carefully constructed to be "highly deceptive" by injecting **local errors** or **global errors**:

- **Local Errors:** The main logic of the option (such as preferred POI categories) matches the user profile, but a preset interference trap is triggered in key details. For example, in the candidate description "*The user has a stable habit of going to mid-to-high-end Jiangsu and Zhejiang cuisine restaurants at noon on workdays... and has multiple actual arrival records,*" the model needs to keenly identify fine-grained errors such as "noon on workdays" (a spatio-temporal displacement: the user actually visits these restaurants on weekend noons, not workday noons)..
- **Global Errors:** The overall attribution of the option completely violates the user’s true decision style, belonging to a forced splicing across scenarios.

Supported by the multiple-choice setting, the model cannot rely on the shortcut of "stopping once a correct option is found," and must make independent true/false judgments for each candidate. To systematically quantify this fine-grained discrimination ability, distractors are designed along the same three orthogonal dimensions as in Task 1, spatio-temporal, profile-evidence, and causal:

- (1) **Spatio-temporal Validity Trap:** Intentionally displacing the spatio-temporal window where the behavior occurred (such as shifting a weekend family dinner to a workday noon), or reviving a dormant historical scenario. This trap requires the model not only to perform static preference matching but also to possess dynamic spatio-temporal joint verification capabilities.
- (2) **Intent-Fulfillment Confusion Trap:** Tampering with the evidence confidence of the behavior, e.g., disguising weak intent signals (such as mere browsing or unconverted clicks) as stable preferences with a high arrival weight (such as fabricating "multiple actual arrival records"), testing the model's precise calibration ability for evidence strength.
- (3) **Contextual Attribution Trap:** Introducing explanations that violate the user's inherent decision style, invert preference priorities, or misread the true motivation (i.e., global errors). For example, citing a one-off five-star hotel stay during the user's out-of-town trip to incorrectly infer an upward or downward shift in the user's everyday local dining tier, testing the model's complex causal reasoning and common-sense generalization.

Step 3: Ground Truth Design and Review. Same as Section 3.3.2 Step 3.

3.5 Task 3: General Mobility Tasks

General mobility tasks include seven core tasks to systematically evaluate the capability boundaries of large models in spatio-temporal understanding and logical reasoning.

3.5.1 POI Semantic Understanding. This task is independent of trajectory data and aims to evaluate the model's mastery of static world knowledge of geographic entities. It includes two directions: forward inference is split into two subtasks: (i) POI→Region, predicting the province, city, and district given a POI name; and (ii) POI→Tag, predicting the category tags. Backward inference (Desc.→POI) requires the model to infer the 5 most likely POI names from a descriptive text. Based on the knowledge base ground truth, forward inference for province/city/district matching uses exact match Accuracy, and tag matching uses F1 Score; backward inference uses Hit@5 for exact candidate matching.

3.5.2 Trajectory Fact Retrieval. Evaluates the model's ability to accurately locate and extract facts from long-context time-series data. It contains three sub-tasks: querying the POI where the user is located at a given specific time (time localization), querying all stay start and end time periods for a given POI (time period extraction), and calculating the "most interested POI" and specific frequency with the highest occurrence frequency among all behaviors within a given time window. This task greatly tests the model's precise counting and matching capabilities in long sequences. Both time localization and time period extraction use exact match Accuracy,

while the most interested POI uses Joint Accuracy, requiring both the POI and frequency to be completely correct at the same time.

3.5.3 Next-Step Mobility Prediction. A classic Next POI prediction task in recommendation systems and LBS. We randomly extract a ground truth from the user's real mobility trajectory for truncation. The model needs to receive the historical trajectory before the truncation point and the current state (time, location) at the time of truncation, and predict the user's next mobility transportation mode, destination POI, and its category tags in a Q&A format. To improve the feasibility of prediction, we require that the ground truth must have appeared in the historical trajectory, using exact match Accuracy.

3.5.4 Location-Time Reasoning. Evaluates the model's bidirectional logical reasoning capabilities across both time and space dimensions. Since it is difficult for the model to provide precise results for time intervals, we require the model to complete two reasoning tasks in a 4-way multiple-choice format: predicting the user's most likely location based on a given post-truncation time, or conversely predicting the start and end time interval of the user's first stay based on a given post-truncation target POI. Similarly, the ground truth is taken from the complete real trajectory, and distractors are sampled from other behaviors in the trajectory. We adopt exact-match Accuracy on the multiple-choice options..

3.5.5 Mobility Preference Summarization. Tests the model's ability to macroscopically induce user behavior characteristics across single events and from long-term complete sequences. Through Q&A, the model is required to summarize the user's overall travel mode preferences in their usual place of residence, as well as subdivided travel mode preferences for specific types of destinations (e.g., given the "Hotel" category). The ground truth is generated by an independent long-term user profile system and corresponds to the mobility trajectory. We report classification Accuracy as the metric.

3.5.6 Mobility Reason Inference. A deep attribution and multi-label classification task. Given a user's historical trajectory and a specific target mobility behavior (including time, destination, travel mode), the model needs to analyze its deep motivation and select all logically consistent reason combinations from a candidate set (commuting, short-term intent, frequently visited place, exploration). The ground truth is calculated from the real mobility trajectory based on the definition of each reason; for example, "exploration" refers to a place the user has never visited before. The metric uses exact match Accuracy for multiple-choice options.

3.5.7 Counterfactual Anomaly Detection. Tests the model's spatio-temporal physical common sense and anti-hallucination capabilities. We extract 20 continuous event segments and inject subtle unreasonable tampering into a certain behavior within the segment (such as long-distance cycling in extreme weather, staying at commercial venues in the early morning, or time sequence contradictions) to generate a tampered version, which is then shuffled and mixed with the original reasonable segment for display. The evaluated model needs to judge which of the two segments is more reasonable and point out the exact index of the unreasonable anomalous record

in the other segment. The Accuracy of rationality judgment and anomaly localization are calculated separately.

4 Experiments

4.1 Experimental Setup

4.1.1 Evaluated Models. To comprehensively evaluate LLMs in LBS scenarios, we select a diverse set of models covering different architectures and scales, including both closed-source and open-source LLMs.

Closed-source models include representative state-of-the-art models such as GPT-series (GPT-4.1 [20], GPT-5.4 [21]), Gemini-series (Gemini-3.1-Pro-Preview [12], Gemini-3.1-Flash-Lite-Preview [11]), Claude-series (Claude-Opus-4.6 [1], Claude-Sonnet-4.6 [2]), and Qwen3.6-Plus [23]. For open-source models, we adopt the Qwen3.5 series [22] (Qwen3.5-27B, Qwen3.5-35B-A3B, Qwen3.5-122B-A10B), DeepSeek-V3.2 [7], Kimi-K2.5 [26], and MiniMax-M2.5, including both dense and Mixture-of-Experts (MoE) architectures.

For fair comparison, we set the temperature to 0.7 and the maximum output length to 32,768 tokens for all evaluated models. The Qwen3.5 series has thinking mode enabled.

4.1.2 Dataset Statistics. To ensure statistical robustness and task diversity, we construct dedicated evaluation sets for each task.

Specifically, the contextual constraint inference dataset contains 2,148 instances (484 single-choice and 1,664 multiple-choice), while the mobility intent inference dataset consists of 1,888 instances. For general mobility tasks, the POI semantic understanding task includes 1,000 POI samples. The trajectory fact retrieval is constructed from 1,000 users, with three questions. The next-step mobility prediction (1,000 users), location-time reasoning (899 users), mobility preference summarization (1,156 users) and counterfactual anomaly detection (1,341 users) tasks each include two questions per user. The mobility reason inference task contains 784 instances.

4.2 Main Results

As shown in Table 3 and Table 4, our extensive evaluation across Mobility Intent Inference (MII), Contextual Constraint Inference (CCI), and General Mobility Tasks (GMT) yields several critical insights. First, a distinct capability hierarchy exists among current LLMs. Leading closed-source models, particularly Gemini-3.1-Pro-Preview and Claude-Opus-4.6, establish the state-of-the-art across most dimensions. However, top-tier open-source models, notably the Qwen series (such as Qwen3.5-35B-A3B), demonstrate highly competitive performance, occasionally surpassing closed-source models in specific foundational mobility tasks. This finding highlights the practical potential of open-source mobility intelligence and points to a promising path toward efficient, privacy-preserving deployment through future model compression and edge-side adaptation.

Second, the fine-grained analysis of MII and CCI tasks exposes significant vulnerabilities in current models when handling complex causal and spatio-temporal traps. While most models achieve over 90% accuracy on single-choice constraint verification, their performance degrades precipitously in multi-choice scenarios. Specifically, models struggle to distinguish true decision logic from deceptive distractors involving temporal-misalignment and deep-conflict traps.

Although Claude-Opus-4.6 exhibits superior robustness against these traps, the overall performance gap indicates that reliable implicit intent reasoning under multi-source constraints remains a formidable challenge.

Finally, the general mobility tasks reveal a clear gap between models’ basic factual perception and their higher-level spatio-temporal reasoning. While LLMs perform strongly on fundamental trajectory fact retrieval tasks, such as temporal localization, they exhibit substantial bottlenecks on tasks that require long-context frequency counting (e.g., interest POI retrieval), dynamic next-step prediction (with accuracy generally below 40%), and precise counterfactual anomaly detection. These findings suggest that the main limitation of current LLMs in LBS is not a lack of static geographic knowledge, but rather the inability to sustain precise, physically grounded, and temporally consistent reasoning over long-horizon behavioral sequences. We hope LBS-IntentBench will encourage future research on models that more effectively integrate causal reasoning with real-world physical constraints in dynamic mobility environments.

4.2.1 Results in Mobility Intent Inference. On MII, Gemini-3.1-Pro-Preview achieves the best overall performance, while Qwen3.6-Plus, Claude-Opus-4.6, and Qwen3.5-35B-A3B also form a strong tier. The gap between Top-1 Accuracy and Exact Match suggests that ranking secondary intents remains challenging. Across models, temporal-misalignment and deep-conflict distractors are among the hardest cases, indicating limited robustness to outdated or causally inconsistent scenarios.

4.2.2 Results in Contextual Constraint Inference. CCI reveals a sharp difference between identifying one plausible explanation and validating all valid explanations. Single-choice accuracy is high for most models, but multi-choice accuracy is substantially lower. This indicates that current LLMs still struggle to enforce precise decision boundaries when multiple constraints interact. Claude-Opus-4.6 performs best overall, and open-source models narrow the gap more on CCI than on MII.

4.2.3 Results in General Mobility Tasks. Gemini-3.1-Pro-Preview achieves the best performance across most tasks, with clear advantages in fact retrieval and counterfactual anomaly detection; Claude-Opus-4.6 follows closely, showing strong performance in mobility preference summarization and location-time reasoning. Overall, leading closed-source LLMs still maintain a noticeable advantage on these tasks. In contrast, GPT-5.4 struggles in fact retrieval, primarily due to limitations in counting ability. The Qwen3.5 series demonstrates strong counting performance, where even smaller-scale variants outperform several top closed-source models.

Trajectory Length. In the trajectory fact retrieval task, most models perform poorly on Q3 (the most interested POI and its count). Prior studies have shown that LLMs exhibit systematic weaknesses in basic counting [8], and that long contexts further amplify errors in retrieval and reasoning [31]. To investigate this phenomenon, we construct two additional datasets with shorter behavior sequences—50 events and 100 events. The results in Figure 2 show that shortening the sequence consistently improves Joint Accuracy across models, indicating that long sequences indeed increase the difficulty of frequency counting and POI identification. The GPT series, Kimi, and Gemini-3.1-Flash-Lite-Preview fail to achieve

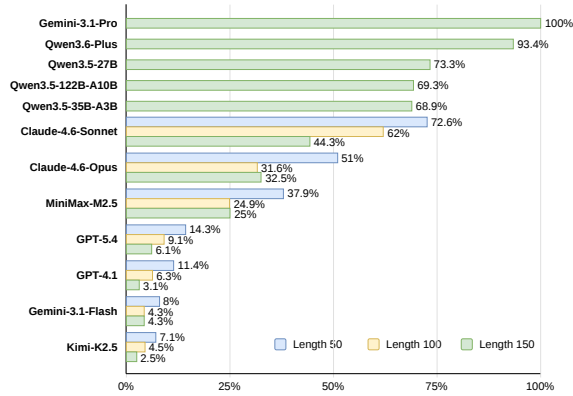
Table 3: Results on Mobility Intent Inference (MII) and Contextual Constraint Inference (CCI)

Model	Mobility Intent Inference (MII)					Contextual Constraint Inference (CCI)					
	Exact Match (%)	Top-1 Acc. (%)	Temp. Mis. (%)	Prof. Viol. (%)	Deep Conf. (%)	Overall (%)	Single (%)	Multi (%)	Temp. Mis. (%)	Prof. Viol. (%)	Deep Conf. (%)
<i>Closed-Source Models</i>											
Gemini-3.1-Pro-Preview	59.89	79.73	84.2	85.7	82.4	66.49	91.5	59.0	74.9	77.9	61.9
Gemini-3.1-Flash-Lite-Preview	46.56	66.05	52.6	57.1	64.7	49.00	91.3	36.7	12.7	18.3	26.9
GPT-5.4	43.13	66.43	57.9	64.3	64.7	54.30	96.7	42.0	36.0	45.5	43.4
GPT-4.1	46.55	68.50	63.2	71.4	70.6	40.21	90.1	25.7	2.3	7.2	21.1
Claude-Opus-4.6	52.17	76.54	68.4	78.6	82.4	73.26	96.9	66.3	83.8	87.8	72.2
Claude-Sonnet-4.6	42.74	70.82	57.9	64.3	70.6	51.14	94.2	38.6	37.5	39.1	31.5
Qwen3.6-Plus	56.89	76.22	78.9	78.6	76.5	63.08	93.8	54.1	66.5	69.2	55.9
<i>Open-Source Models</i>											
Kimi-K2.5	47.19	68.75	68.4	71.4	70.6	59.92	95.5	49.6	55.1	61.7	48.0
MiniMax-M2.5	55.26	71.68	73.7	71.4	76.5	51.28	90.1	39.4	33.4	38.2	35.7
DeepSeek-V3.2	37.82	60.12	52.6	57.1	58.8	49.77	91.9	37.5	15.1	28.5	29.8
Qwen3.5-122B-A10B	33.01	65.41	42.1	50.0	47.1	58.75	92.1	49.4	58.8	58.3	46.9
Qwen3.5-35B-A3B	55.40	75.48	73.7	78.6	76.5	58.05	93.4	47.8	56.6	55.3	46.2
Qwen3.5-27B	44.15	72.37	63.2	64.3	67.6	60.47	94.8	50.5	61.4	67.2	51.3

Table 4: Results on General Mobility Tasks with seven subtasks

Model	POI Semantic Understanding			Trajectory Fact Retrieval			Next-Step Mobility Prediction		Location-Time Reasoning		Mobility Preference Summarization		Mobility Reason Inference	Counterfactual Anomaly Detection	
	Acc (%)	F1 (%)	Hit@5 (%)	Acc (%)	Acc (%)	Joint Acc (%)	Acc (%)	Acc (%)	Acc (%)	Acc (%)	Acc (%)	Acc (%)	EM (%)	Acc (%)	Acc (%)
	(POI→Region)	(POI→Tag)	(Desc→POI)	(Time Loc.)	(Stay Seg.)	(Interest POI)	(Dest. POI)	(Dest. Tag)	(Loc. Pred.)	(Time Pred.)	(Overall Pref.)	(Category Pref.)	(Multi-choice)	(Plausibility)	(Anomaly Loc.)
Closed-Source Models															
Gemini-3.1-Pro-Preview	72.1	60.3	41.2	100.0	99.9	100.0	35.4	40.6	82.0	67.0	77.3	66.9	59.6	83.4	70.3
Gemini-3.1-Flash-Lite-Preview	64.6	58.3	29.3	96.0	90.7	4.3	28.5	31.1	77.6	55.1	78.6	64.9	36.1	58.5	23.6
GPT-5.4	66.3	63.2	37.6	96.0	98.7	6.1	29.3	33.4	78.9	59.0	79.4	66.4	41.1	71.5	49.0
GPT-4.1	52.7	46.7	29.2	87.3	90.2	3.1	24.2	28.5	70.1	56.1	78.3	63.8	28.4	54.3	22.1
Claude-Opus-4.6	64.3	56.3	37.5	99.4	97.3	32.5	33.4	36.4	82.9	62.3	79.8	69.7	55.0	76.7	59.3
Claude-Sonnet-4.6	58.4	53.6	35.7	98.4	97.7	44.3	30.4	34.8	78.6	59.3	77.8	61.9	56.6	70.0	50.4
Qwen3.6-Plus	67.8	54.9	35.9	99.5	98.7	93.4	29.5	33.7	82.2	59.8	79.7	69.6	54.2	71.9	42.9
Open-Source Models															
Kimi-K2.5	65.1	54.6	35.5	92.3	88.5	2.5	30.8	33.4	79.9	56.7	78.6	64.8	51.4	61.8	29.8
MiniMax-M2.5	59.5	55.3	34.1	99.6	92.1	25.0	26.5	29.5	71.1	55.2	76.6	64.6	49.3	60.1	30.3
DeepSeek-V3.2	65.8	50.3	35.1	91.2	93.3	2.8	27.1	30.8	71.0	56.2	77.9	67.2	36.6	63.6	32.3
Qwen3.5-122B-A10B	58.6	53.3	33.5	99.8	97.7	69.3	21.1	34.0	76.1	60.0	75.8	66.4	56.8	62.3	24.8
Qwen3.5-35B-A3B	54.7	55.4	32.8	99.5	98.4	68.9	23.3	34.9	75.0	61.6	77.9	68.3	59.0	63.1	22.2
Qwen3.5-27B	51.9	51.9	32.8	99.9	99.3	73.3	22.3	34.8	77.3	58.3	77.4	67.3	58.0	63.3	27.4

accurate counting even on short sequences, suggesting more fundamental limitations in frequency counting.

**Figure 2: Performance of different models on the trajectory fact retrieval task (Q3) under varying sequence lengths.**

5 Conclusion

In this paper, we introduce LBS-IntentBench, a benchmark for implicit intent inference and spatio-temporal reasoning in LLM-based location recommendation. It is grounded in large-scale desensitized real-user trajectories and organized around an Intent–Decision–Fact hierarchy comprising three complementary tasks: Mobility Intent

Inference (ranking concurrent latent intents under query-free spatio-temporal triggers), Contextual Constraint Inference (identifying decision logic that jointly satisfies temporal, profile, and contextual constraints), and General Mobility Tasks (covering seven subtasks from POI semantic understanding and trajectory fact retrieval to next-step mobility prediction, preference induction, and counterfactual anomaly detection). To enable fine-grained diagnosis, we inject three systematic distractor categories—temporal misalignment, profile constraint violation, and causal inversion—and pair task-specific metrics (Exact Match, Top-1 Acc, Hit@5) with LLM-generated questions curated under an 80% expert-review consensus threshold. Experiments across thirteen LLMs show highly imbalanced capability profiles: models are broadly competent on static geographic knowledge and local fact retrieval, yet remain far from solving implicit intent ranking, multi-source constraint composition, and long-horizon structured aggregation, suggesting that physically grounded LLM-based recommendation in LBS remains far from solved.

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